

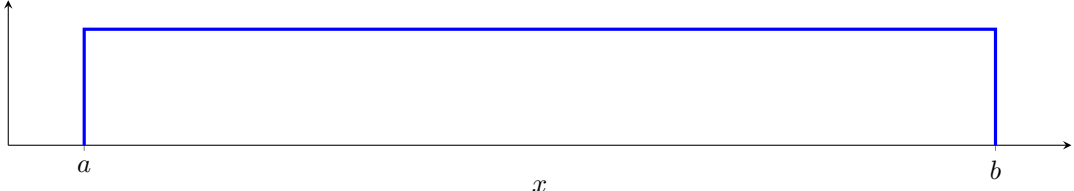
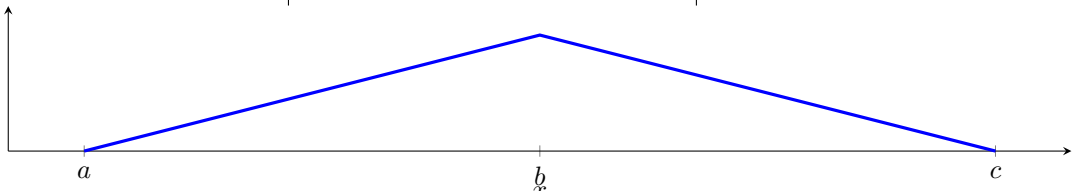
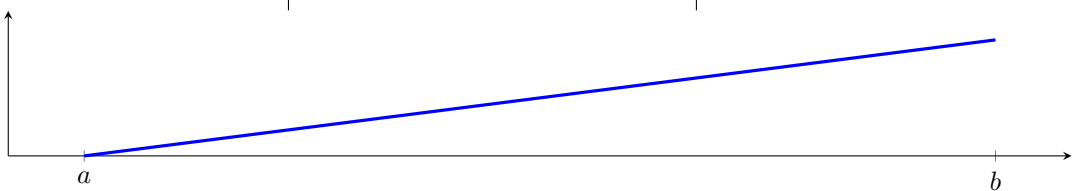
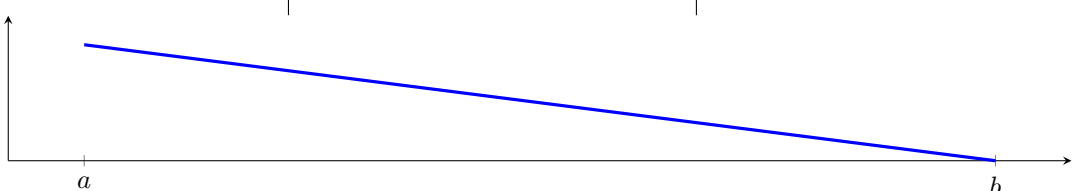
Practice: Mean and Variance of Continuous Distributions

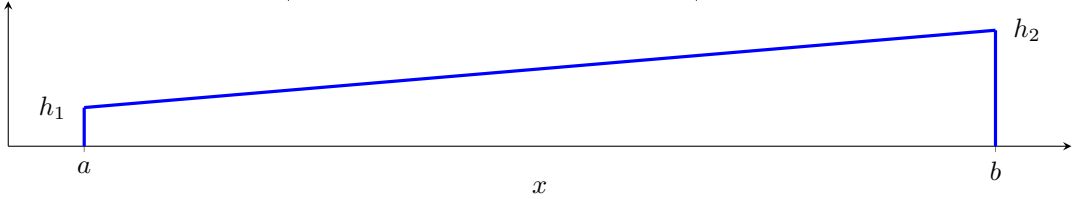
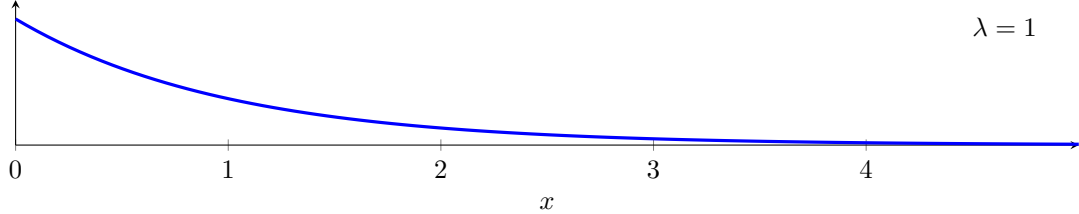
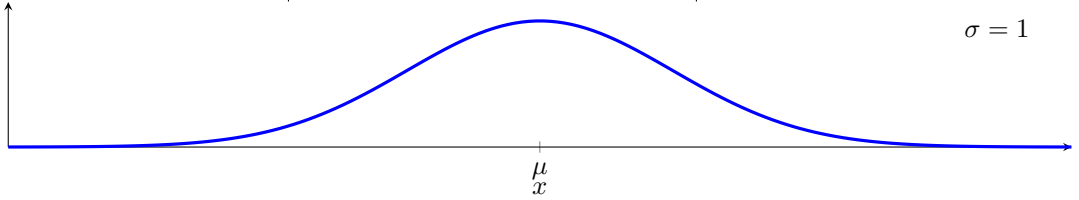
Purpose. Calculate and interpret the mean (expected value) and variance of continuous probability distributions using standard formulas.

Instructions. For each situation:

1. Identify the probability distribution and its parameters.
2. Calculate the mean, or expected value, $E(X)$.
3. Calculate the variance, $Var(X)$.
4. Calculate the standard deviation, $\sigma_X = \sqrt{Var(X)}$.

Useful formulas

Distribution	Notation	$E(X)$	$Var(X)$
Uniform	$X \sim U(a, b)$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
			
Triangular	$X \sim \text{Tri}(a, b, c)$	$\frac{a + b + c}{3}$	$\frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}$
			
Linear increasing	$X \sim \text{Lin}_{\text{inc}}(a, b)$	$\frac{a + 2b}{3}$	$\frac{(b - a)^2}{18}$
			
Linear decreasing	$X \sim \text{Lin}_{\text{dec}}(a, b)$	$\frac{2a + b}{3}$	$\frac{(b - a)^2}{18}$
			

Distribution	Notation	$E(X)$	$Var(X)$
Trapezoidal	$X \sim \text{Trap}(a, b, h_1, h_2)$	$\frac{b-a}{3} \frac{h_1^2 + h_1 h_2 + h_2^2}{h_1 + h_2} + \frac{a+b}{2}$	$\frac{(b-a)^2}{18} \frac{h_1^2 + 2h_1 h_2 + h_2^2}{(h_1 + h_2)^2}$
			
Exponential	$X \sim \text{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
			
Normal	$X \sim N(\mu, \sigma^2)$	μ	σ^2
			

Distribution	Notation	$E(X)$	$Var(X)$
Uniform	$X \sim U(a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Triangular	$X \sim \text{Tri}(a, b, c)$	$\frac{a+b+c}{3}$	$\frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}$
Linear increasing	$X \sim \text{Lin}_{\text{inc}}(a, b)$	$\frac{a+2b}{3}$	$\frac{(b-a)^2}{18}$
Linear decreasing	$X \sim \text{Lin}_{\text{dec}}(a, b)$	$\frac{2a+b}{3}$	$\frac{(b-a)^2}{18}$
Trapezoidal	$X \sim \text{Trap}(a, b, h_1, h_2)$	$\frac{b-a}{3} \frac{h_1^2 + h_1 h_2 + h_2^2}{h_1 + h_2} + \frac{a+b}{2}$	$\frac{(b-a)^2}{18} \frac{h_1^2 + 2h_1 h_2 + h_2^2}{(h_1 + h_2)^2}$
Exponential	$X \sim \text{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	$X \sim N(\mu, \sigma^2)$	μ	σ^2

1. Waiting for a bus after school.

A student waits for a bus after school. The waiting time can be anywhere between 2 and 10 minutes, and all waiting times in this interval are equally likely.

- a) Identify the distribution and its parameters.
- b) Calculate the mean waiting time.
- c) Calculate the variance of the waiting time.
- d) Calculate the standard deviation of the waiting time.
- e) Calculate the probability that the waiting time is within one standard deviation of the mean.

2. Delivery time for a small business.

A local business records the time required to deliver an order within the city. Delivery times range from 20 to 50 minutes. The probability density increases linearly from 20 to 50 minutes, so longer delivery times are more likely than shorter ones.

- a) Identify the distribution and its parameters.
- b) Calculate the mean delivery time.
- c) Calculate the variance of the delivery time.
- d) Calculate the standard deviation of the delivery time.
- e) Calculate the probability that the delivery time is within one standard deviation of the mean.

3. Daily sales at a school café.

A school café records the number of sandwiches sold during the lunch period. On a typical day, sales range from 20 to 80 sandwiches, with 50 sandwiches being the most common result. The probability density increases linearly from 20 to 50 and decreases linearly from 50 to 80.

- a) Identify the distribution and its parameters.
- b) Calculate the mean number of sandwiches sold.
- c) Calculate the variance of the number of sandwiches sold.
- d) Calculate the standard deviation of the number of sandwiches sold.
- e) Calculate the probability that the number of sandwiches sold is within one standard deviation of the mean.

4. Time spent in a school library.

A student visits the school library after classes. The time spent in the library ranges from 20 to 80 minutes. Shorter visits are more common than longer visits, and the probability density decreases linearly from 20 to 80 minutes.

- a) Identify the distribution and its parameters.
- b) Calculate the mean time spent in the library.
- c) Calculate the variance of the time spent in the library.
- d) Calculate the standard deviation of the time spent in the library.
- e) Calculate the probability that the time spent in the library is within one standard deviation of the mean.

5. Time spent at a university career fair.

A visitor attends a career fair. The time spent at the event ranges from 0 to 6 hours. The probability density increases linearly from 0 to 2 hours, remains constant from 2 to 4 hours, and decreases linearly from 4 to 6 hours until it reaches zero.

- a) Identify the distribution and its parameters.
- b) Calculate the mean time spent at the fair.
- c) Calculate the variance of the time spent at the fair.
- d) Calculate the standard deviation of the time spent at the fair.
- e) Calculate the probability that the time spent at the fair is within one standard deviation of the mean.

6. Time between customers at a bakery.

A bakery records the time between consecutive customers during a busy period. The time between arrivals is modeled by an exponential distribution. On average, one customer arrives every 4 minutes.

- a) Identify the parameter λ .
- b) Calculate the mean time between arrivals.
- c) Calculate the variance of the time between arrivals.
- d) Calculate the standard deviation of the time between arrivals.
- e) Calculate the probability that the time between arrivals is within one standard deviation of the mean.

7. Travel time to school.

The daily travel time for students commuting to school is modeled by a normal distribution with a mean of 45 minutes and a standard deviation of 8 minutes.

- a) Identify the parameters of the normal distribution.
- b) Calculate the mean travel time.
- c) Calculate the variance of the travel time.
- d) Calculate the standard deviation of the travel time.
- e) Calculate the probability that the travel time is within one standard deviation of the mean.

Solutions

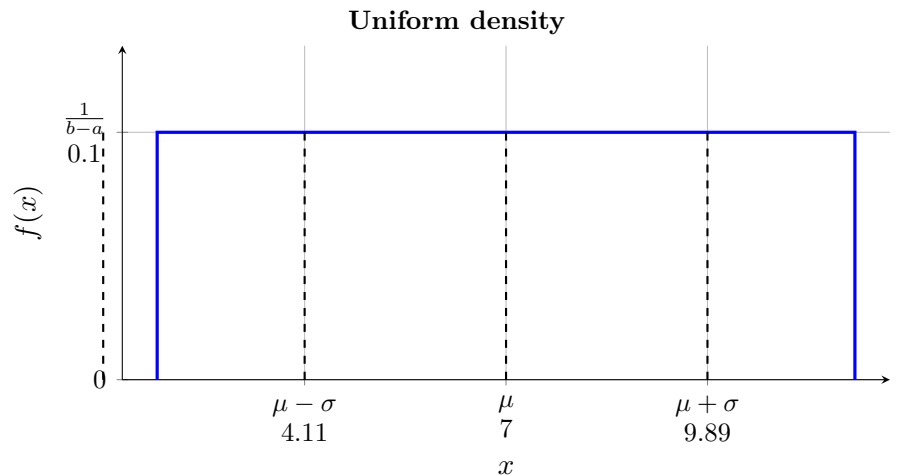
1 Uniform distribution

1.1 Distribution and parameters

A small business invests in a short-term opportunity. Based on similar investments, the money earned is expected to be between \$2,000 and \$12,000, with all values in this range considered equally likely.

Let X represent the money earned, measured in thousands of dollars. Therefore,

$$X \sim U(a, b) \quad \longrightarrow \quad X \sim U(2, 12) \quad \longrightarrow \quad a = 2, \quad b = 12.$$



1.2 Mean

For a uniform distribution,

$$E(X) = \frac{a+b}{2} \quad \longrightarrow \quad E(X) = \frac{2+12}{2} = 7 \quad \longrightarrow \quad E(X) = 7 \text{ thousand dollars}$$

1.3 Variance

For a uniform distribution,

$$Var(X) = \frac{(b-a)^2}{12} \quad \longrightarrow \quad Var(X) = \frac{(12-2)^2}{12} = \frac{100}{12} = \frac{25}{3} \approx 8.33 \quad \longrightarrow \quad Var(X) \approx 8.33 \text{ thousand dollars}^2$$

1.4 Standard Deviation

The standard deviation is

$$\sigma_X = \sqrt{\frac{25}{3}} \approx 2.89 \quad \longrightarrow \quad \sigma_X \approx 2.89 \text{ thousand dollars}$$

1.5 Probability within one standard deviation

We want the probability that the money earned is within one standard deviation of the mean:

$$P(\mu - \sigma_X < X < \mu + \sigma_X)$$

Using $\mu = 7$ and $\sigma_X \approx 2.89$,

$$P(7 - 2.89 < X < 7 + 2.89) = P(4.11 < X < 9.89)$$

For a uniform distribution,

$$P(c < X < d) = \frac{d - c}{b - a}$$

Therefore,

$$P(4.11 < X < 9.89) = \frac{9.89 - 4.11}{12 - 2} = \frac{5.78}{10} \approx 0.578$$

$$P(\mu - \sigma_X < X < \mu + \sigma_X) \approx 0.578$$

Thus, approximately 57.8% of the possible investment earnings are within one standard deviation of the mean.

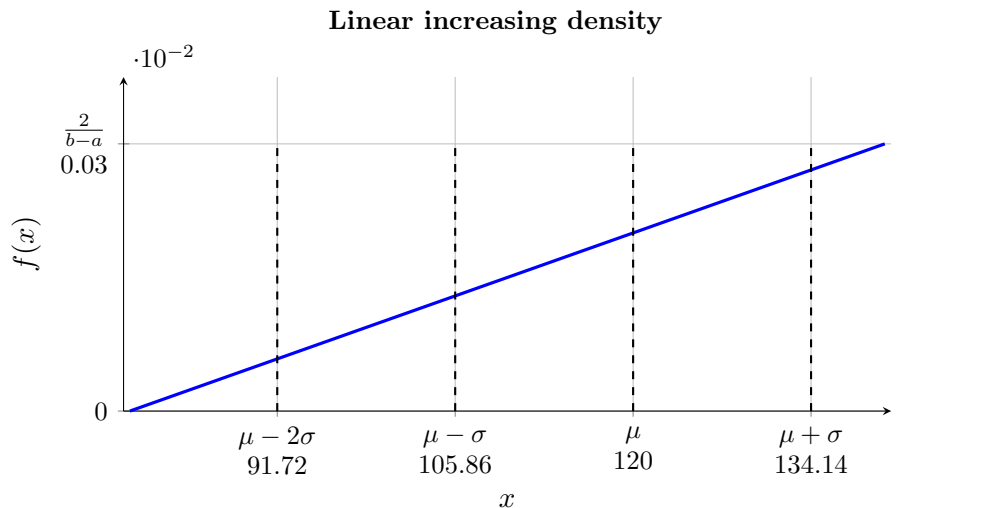
2 Linear increasing distribution

2.1 Distribution and parameters

A bakery prepares custom cakes for weekend orders. Based on previous weekends, the preparation time for a randomly selected cake ranges from 80 to 140 minutes. Longer preparation times are more likely because larger and more detailed cakes require more work, so the probability density increases linearly across this range.

Let T represent the preparation time, in minutes, for one cake. Therefore,

$$T \sim \text{Lin}_{\text{inc}}(a, b) \quad \longrightarrow \quad T \sim \text{Lin}_{\text{inc}}(80, 140) \quad \longrightarrow \quad a = 80 \quad b = 140$$



2.2 Mean

For a linear increasing distribution,

$$E(T) = \frac{a + 2b}{3} \longrightarrow E(T) = \frac{80 + 2(140)}{3} = \frac{360}{3} = 120 \longrightarrow E(T) = 120 \text{ minutes}$$

2.3 Variance

For a linear increasing distribution,

$$Var(T) = \frac{(b - a)^2}{18} \longrightarrow Var(T) = \frac{(140 - 80)^2}{18} = \frac{3600}{18} = 200 \longrightarrow Var(T) = 200 \text{ minutes}^2$$

2.4 Interpretation

The standard deviation is

$$\sigma_T = \sqrt{200} \approx 14.14 \longrightarrow \sigma_T \approx 14.14 \text{ minutes}$$

2.5 Probability within one standard deviation

We want the probability that the preparation time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using $\mu = 120$ and $\sigma_T \approx 14.14$,

$$P(120 - 14.14 < T < 120 + 14.14) = P(105.86 < T < 134.14)$$

For a linear increasing distribution, the probability density is

$$f(t) = \frac{2(t - a)}{(b - a)^2}.$$

Therefore,

$$\begin{aligned} P(105.86 < T < 134.14) &= \int_{105.86}^{134.14} \frac{2(t - 80)}{(140 - 80)^2} dt \\ &= \left[\frac{(t - 80)^2}{3600} \right]_{105.86}^{134.14} \approx 0.627 \end{aligned}$$

$$P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.627$$

Thus, approximately 62.7% of cake preparation times are within one standard deviation of the mean.

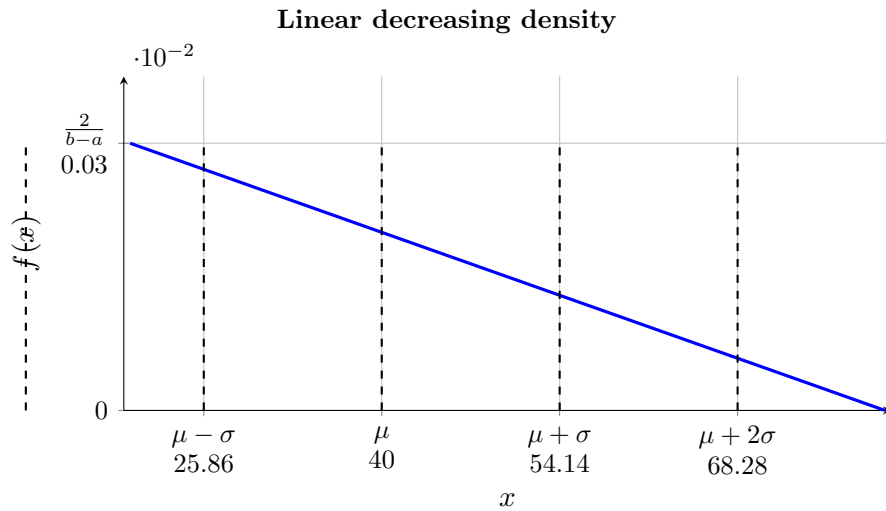
3 Linear decreasing distribution

3.1 Distribution and parameters

A bakery delivers custom cake orders to customers during the weekend. Based on previous deliveries, the delivery time for a randomly selected order ranges from 20 to 80 minutes. Shorter delivery times are more likely because most orders are delivered within the local area, so the probability density decreases linearly as the delivery time increases.

Let T represent the delivery time, in minutes, for one order. Therefore,

$$T \sim \text{Lin}_{\text{dec}}(a, b) \quad \longrightarrow \quad T \sim \text{Lin}_{\text{dec}}(20, 80) \quad \longrightarrow \quad a = 20, \quad b = 80$$



3.2 Mean

For a linear decreasing distribution,

$$E(T) = \frac{2a + b}{3} \quad \longrightarrow \quad E(T) = \frac{2(20) + 80}{3} = \frac{120}{3} = 40 \quad \longrightarrow \quad E(T) = 40 \text{ minutes}$$

3.3 Variance

For a linear decreasing distribution,

$$\text{Var}(T) = \frac{(b - a)^2}{18} \quad \longrightarrow \quad \text{Var}(T) = \frac{(80 - 20)^2}{18} = \frac{3600}{18} = 200 \quad \longrightarrow \quad \text{Var}(T) = 200 \text{ minutes}^2$$

3.4 Interpretation

The standard deviation is

$$\sigma_T = \sqrt{200} \approx 14.14 \quad \longrightarrow \quad \sigma_T \approx 14.14 \text{ minutes}$$

3.5 Probability within one standard deviation

We want the probability that the delivery time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using $\mu = 40$ and $\sigma_T \approx 14.14$,

$$P(40 - 14.14 < T < 40 + 14.14) = P(25.86 < T < 54.14)$$

For a linear decreasing distribution, the probability density is

$$f(t) = \frac{2(b-t)}{(b-a)^2}.$$

Therefore,

$$\begin{aligned} P(25.86 < T < 54.14) &= \int_{25.86}^{54.14} \frac{2(80-t)}{(80-20)^2} dt \\ &= \left[\frac{2(80)t - t^2}{3600} \right]_{25.86}^{54.14} \approx 0.635 \end{aligned}$$

$$P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.635$$

Thus, approximately 63.5% of delivery times are within one standard deviation of the mean.

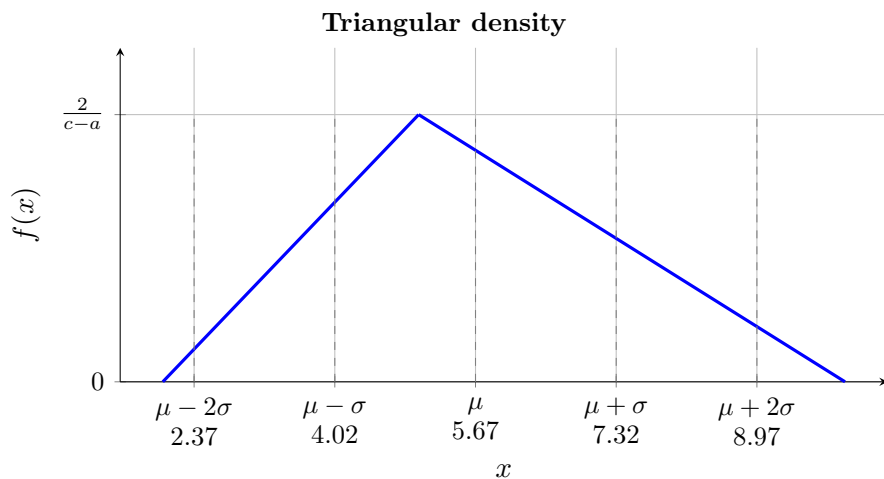
4 Triangular distribution

4.1 Distribution and parameters

A financial institution processes applications for small business loans. Based on previous applications, the processing time ranges from 2 to 10 hours. Most applications can be completed in about 5 hours, while shorter and longer processing times are less common. Therefore, the processing time has a triangular probability density with a mode at 5 hours.

Let T represent the processing time of a loan application, measured in hours. Therefore,

$$T \sim \text{Tri}(a, b, c) \quad \longrightarrow \quad T \sim \text{Tri}(2, 10, 5) \quad \longrightarrow \quad a = 2 \quad b = 10 \quad c = 5$$



4.2 Mean

For a triangular distribution,

$$E(T) = \frac{a + b + c}{3} \longrightarrow E(T) = \frac{2 + 10 + 5}{3} = \frac{17}{3} \approx 5.67 \longrightarrow E(T) \approx 5.67 \text{ hours}$$

4.3 Variance

For a triangular distribution,

$$Var(T) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}.$$

$$\begin{aligned} Var(T) &= \frac{2^2 + 10^2 + 5^2 - (2)(10) - (2)(5) - (10)(5)}{18} \\ \longrightarrow &= \frac{4 + 100 + 25 - 20 - 10 - 50}{18} \longrightarrow Var(T) \approx 2.72 \text{ hours}^2 \\ &= \frac{49}{18} \approx 2.72 \end{aligned}$$

4.4 Interpretation

The standard deviation is

$$\sigma_T = \sqrt{\frac{49}{18}} \approx 1.65 \longrightarrow \sigma_T \approx 1.65 \text{ hours}.$$

4.5 Probability within one standard deviation

We want the probability that the processing time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using $\mu \approx 5.67$ and $\sigma_T \approx 1.65$,

$$P(5.67 - 1.65 < T < 5.67 + 1.65) = P(4.02 < T < 7.32)$$

Since the mode is $c = 5$, the probability density is

$$f(t) = \begin{cases} \frac{2(t-a)}{(b-a)(c-a)}, & a \leq t \leq c, \\ \frac{2(b-t)}{(b-a)(b-c)}, & c < t \leq b. \end{cases}$$

Therefore,

$$\begin{aligned} P(4.02 < T < 7.32) &= \int_{4.02}^5 \frac{2(t-2)}{(10-2)(5-2)} dt + \int_5^{7.32} \frac{2(10-t)}{(10-2)(10-5)} dt \\ &\approx 0.563 \end{aligned}$$

$$P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.563$$

Thus, approximately 56.3% of loan-processing times are within one standard deviation of the mean.

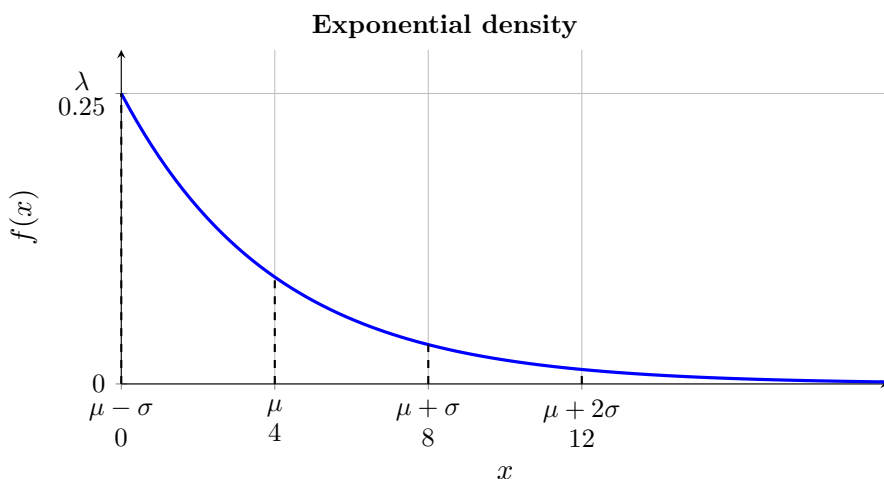
5 Exponential distribution

5.1 Parameter

An online clothing store receives orders throughout the day. Based on its recent sales activity, the store receives an average of 15 orders per hour. Let X represent the time between consecutive orders, measured in minutes.

For an exponential distribution, the rate is the average number of events per unit of time. Since 15 orders occur per hour,

$$\lambda = \frac{15}{60} = \frac{1}{4} \text{ min}^{-1} \quad \longrightarrow \quad \lambda = \frac{1}{4} \text{ min}^{-1}$$



5.2 Mean

For an exponential distribution,

$$E(X) = \frac{1}{\lambda} \quad \longrightarrow \quad E(X) = \frac{1}{1/4} = 4 \quad \longrightarrow \quad E(X) = 4 \text{ minutes}$$

5.3 Variance

For an exponential distribution,

$$Var(X) = \frac{1}{\lambda^2} \quad \longrightarrow \quad Var(X) = \frac{1}{(1/4)^2} = 16 \quad \longrightarrow \quad Var(X) = 16 \text{ minutes}^2$$

5.4 Interpretation

The standard deviation is

$$\sigma_X = \sqrt{16} = 4 \quad \longrightarrow \quad \sigma_X = 4 \text{ minutes}$$

The mean and standard deviation are both 4 minutes, reflecting the characteristic variability of an exponential waiting-time distribution.

5.5 Probability within one standard deviation

We want the probability that the time between consecutive orders is within one standard deviation of the mean:

$$P(\mu - \sigma_X < X < \mu + \sigma_X)$$

Using $\mu = 4$ and $\sigma_X = 4$,

$$P(4 - 4 < X < 4 + 4) = P(0 < X < 8)$$

For an exponential distribution, the cumulative distribution function is

$$P(X < x) = 1 - e^{-\lambda x}.$$

Therefore,

$$P(0 < X < 8) = 1 - e^{-(1/4)(8)} = 1 - e^{-2} \approx 0.8647$$

$$P(\mu - \sigma_X < X < \mu + \sigma_X) \approx 0.865$$

Thus, approximately 86.5% of the time intervals between consecutive orders are within one standard deviation of the mean.

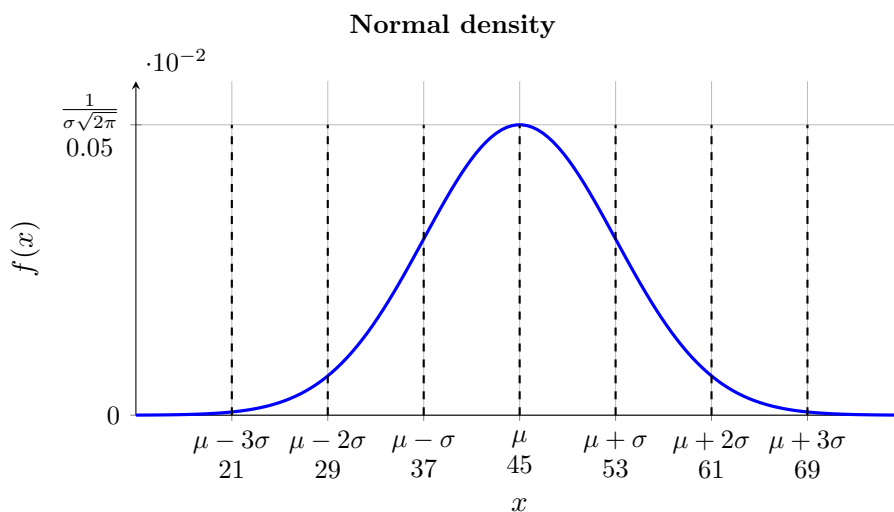
6 Normal distribution

6.1 Parameters

A small retail business tracks its monthly sales revenue over several months. Based on its historical sales data, the monthly revenue is approximately centered at \$45,000, with a standard deviation of \$8,000. Let Y represent the monthly sales revenue, measured in thousands of dollars.

Therefore,

$$Y \sim N(45, 8^2) \quad \longrightarrow \quad \mu = 45 \quad \sigma = 8$$



6.2 Mean

For a normal distribution,

$$E(Y) = \mu \longrightarrow E(Y) = 45 \text{ thousand dollars}$$

6.3 Variance

For a normal distribution,

$$Var(Y) = \sigma^2 \longrightarrow Var(Y) = 8^2 = 64 \longrightarrow Var(Y) = 64 \text{ thousand dollars}^2.$$

6.4 Interpretation

The standard deviation is

$$\sigma_Y = \sqrt{64} = 8 \longrightarrow \sigma_Y = 8 \text{ thousand dollars}.$$

This means that monthly sales revenue typically varies by about \$8,000 around the mean revenue of \$45,000.

6.5 Probability within one standard deviation

We want the probability that monthly sales revenue is within one standard deviation of the mean:

$$P(\mu - \sigma_Y < Y < \mu + \sigma_Y)$$

Using $\mu = 45$ and $\sigma_Y = 8$,

$$P(45 - 8 < Y < 45 + 8) = P(37 < Y < 53)$$

Standardizing the two endpoints,

$$z_1 = \frac{37 - 45}{8} = -1, \quad z_2 = \frac{53 - 45}{8} = 1$$

Therefore,

$$P(37 < Y < 53) = P(-1 < Z < 1) \approx 0.6827$$

$$P(\mu - \sigma_Y < Y < \mu + \sigma_Y) \approx 0.683$$

Thus, approximately 68.3% of monthly sales revenues are within one standard deviation of the mean.